15-441 Computer Networking

Lecture 7 – Bridging, Addressing and Forwarding

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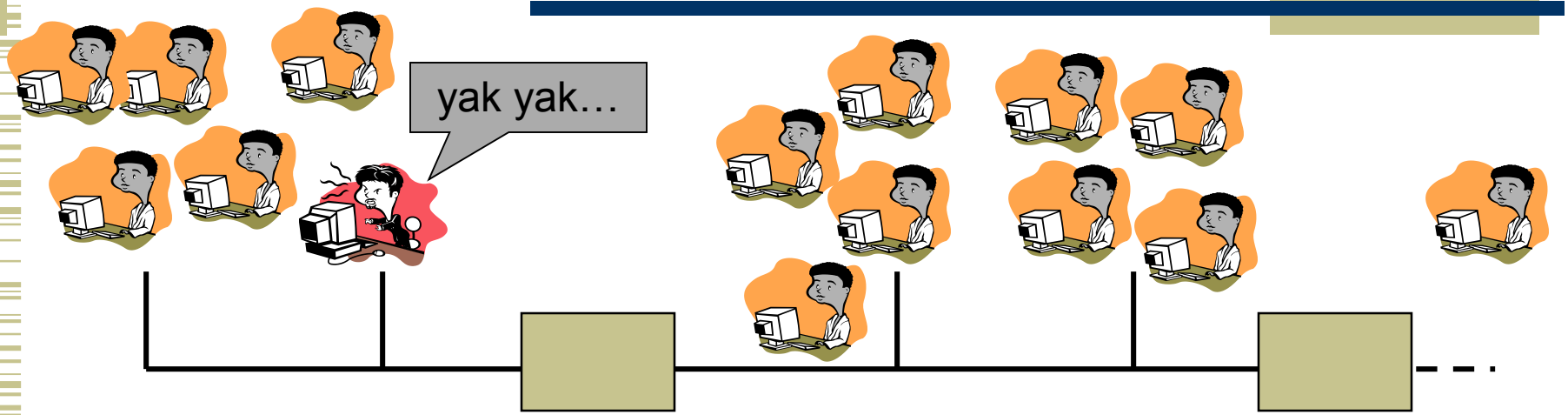
Fall 2013

www.cs.cmu.edu/~prs/15-441-F13

Scale

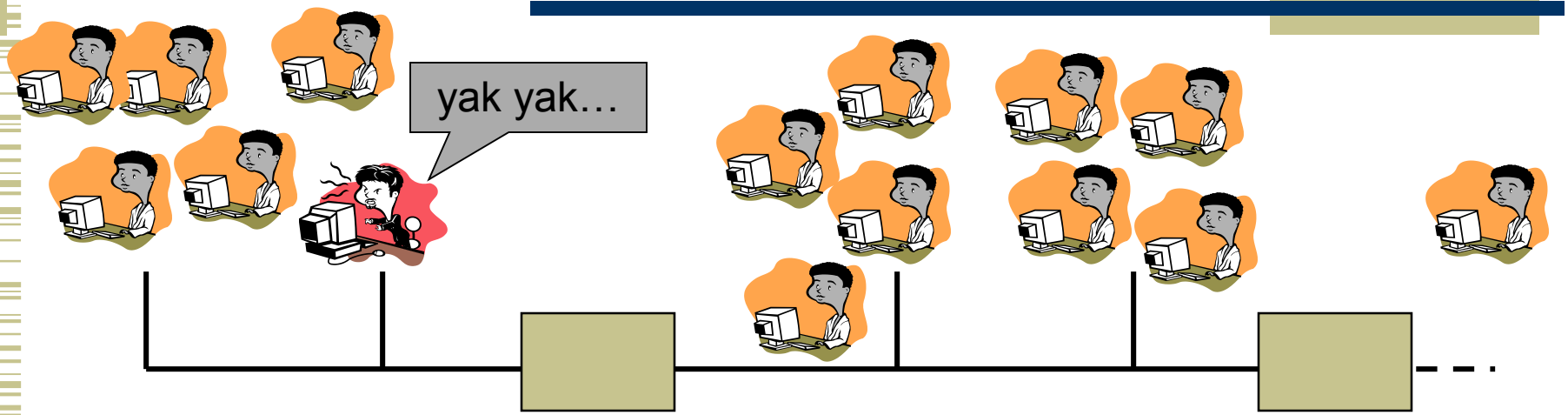
- What breaks when we keep adding people to the same wire?

Scale



- What breaks when we keep adding people to the same wire?
- Only solution: split up the people onto multiple wires
  - But how can they talk to each other?

Problem 1 – Reconnecting LANs



- When should these boxes forward packets between wires?
- How do you specify a destination?
- How does your packet find its way?

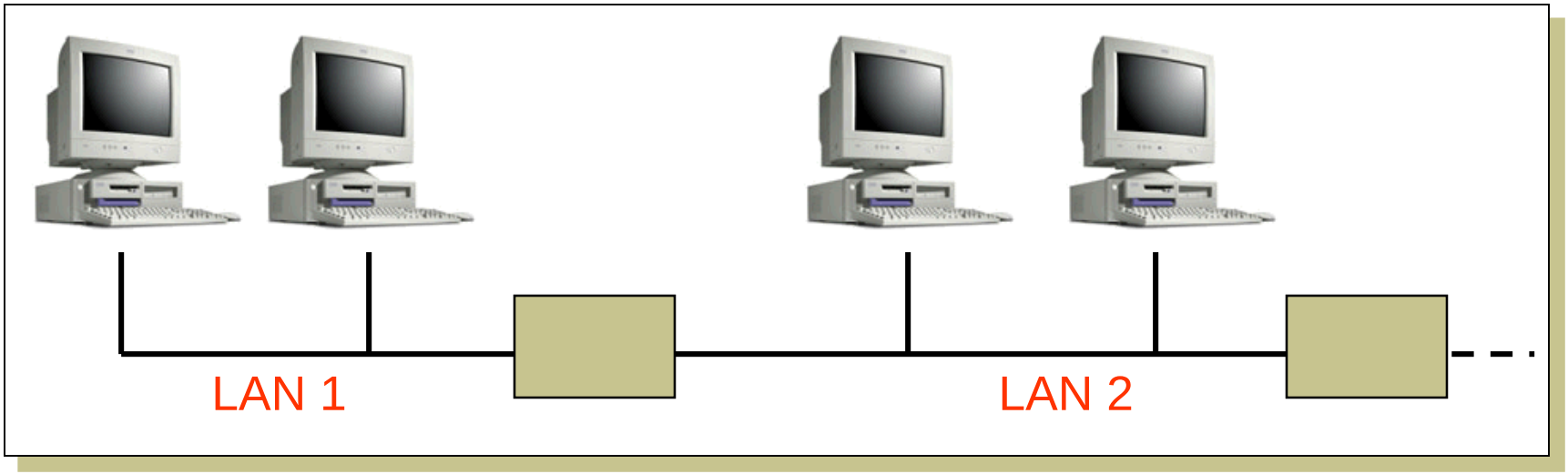
Outline

- Bridging
- Switch Topologies
- Internetworks
  - Methods for packet forwarding
- Traditional IP addressing

Building Larger LANs: Bridges

- Extend reach of a single shared medium
- Connect two or more “segments” by copying data frames between them
  - Only copy data when needed → key difference from repeaters/hubs
  - Reduce collision domain compared with single LAN
  - Separate segments can send at once → much greater bandwidth

- Challenge: learning which packets to copy across links

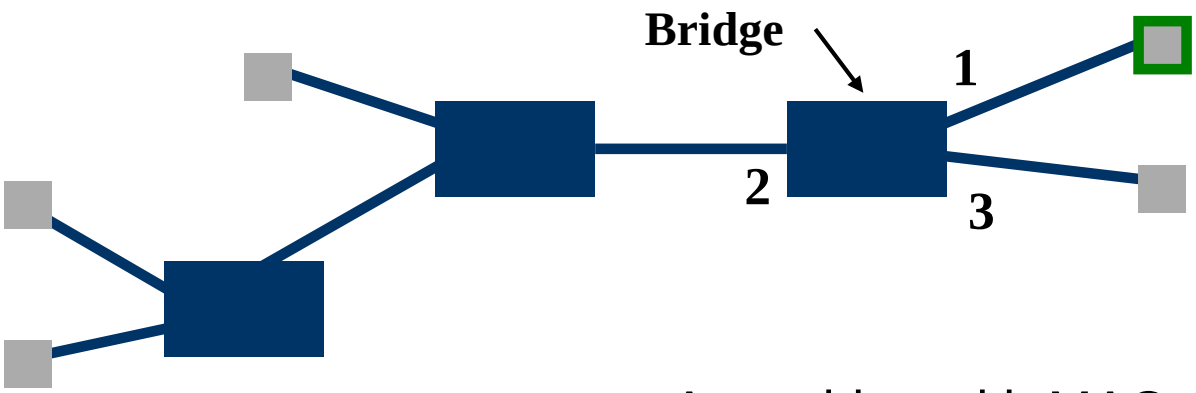


Transparent Bridges

- Design goals:
  - Self-configuring without hardware or software changes
  - Bridge do not impact the operation of the individual LANs

- Three parts to making bridges transparent:
  1) Forwarding frames
  2) Learning addresses/host locations
  3) Spanning tree algorithm

Frame Forwarding

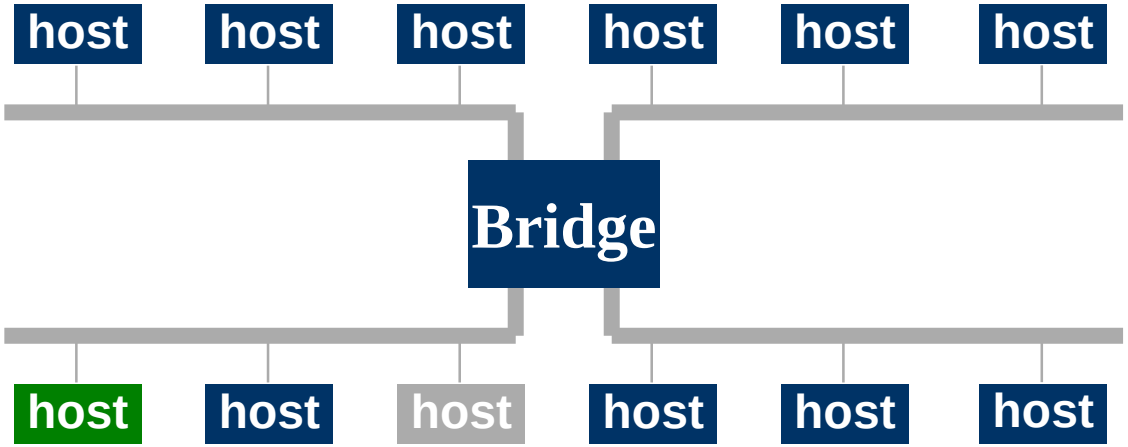


MAC Address	Port	Age
A21032C9A591	1	36
99A323C90842	2	01
8711C98900AA	2	15
301B2369011C	2	16
695519001190	3	11

- A machine with MAC Address lies in the direction of number port of the bridge
- For every packet, the bridge “looks up” the entry for the packets destination MAC address and forwards the packet on that port.
  - Other packets are broadcast – why?
- Timer is used to flush old entries

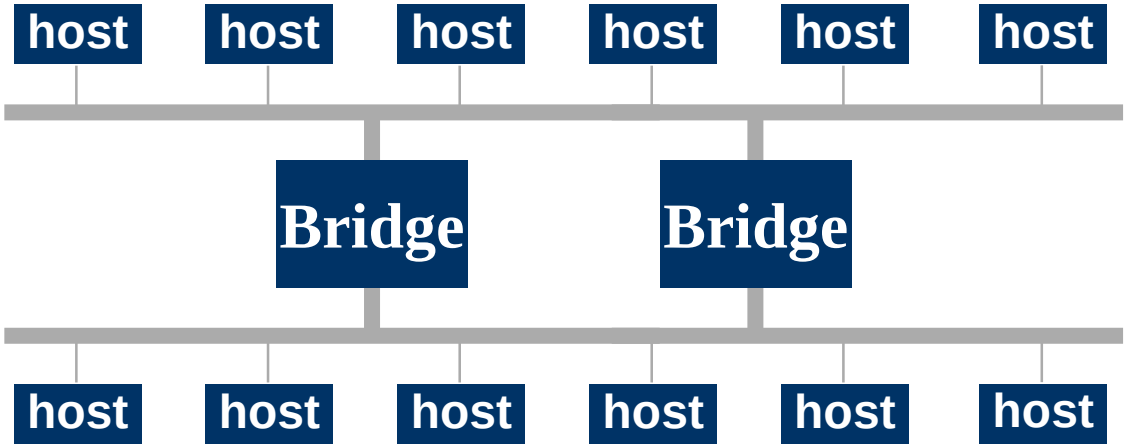
Learning Bridges

- Manually filling in bridge tables?
  - Time consuming, error-prone
- Keep track of source address of packets arriving on every link, showing what segment hosts are on
  - Fill in the forwarding table based on this information



Spanning Tree Bridges

- More complex topologies can provide redundancy.
  - But can also create loops.
- What is the problem with loops?
- Solution: spanning tree



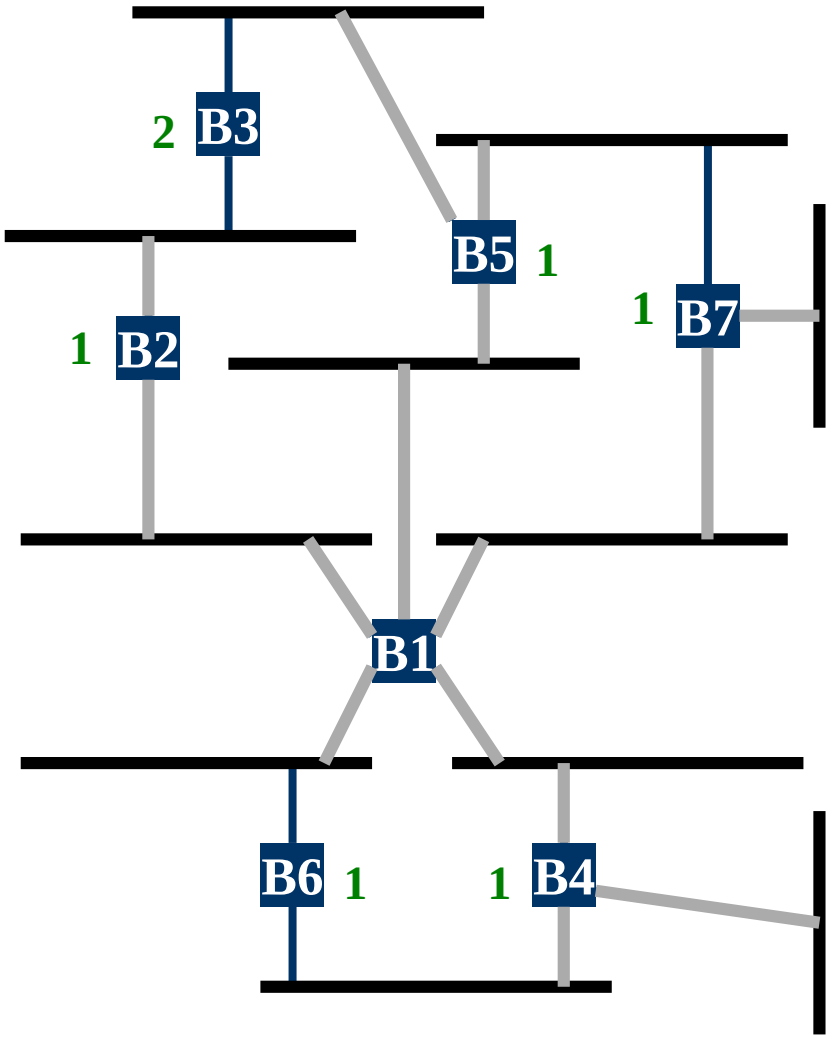
Spanning Tree Protocol Overview

Embed a tree that provides a single unique path to each destination:

1) Elect a single bridge as a root bridge
2) Each bridge calculates the distance of the shortest path to the root bridge
3) Each LAN identifies a *designated bridge*, the bridge closest to the root. It will forward packets to the root.
4) Each bridge determines a *root port*, which will be used to send packets to the root
5) Identify the ports that form the spanning tree

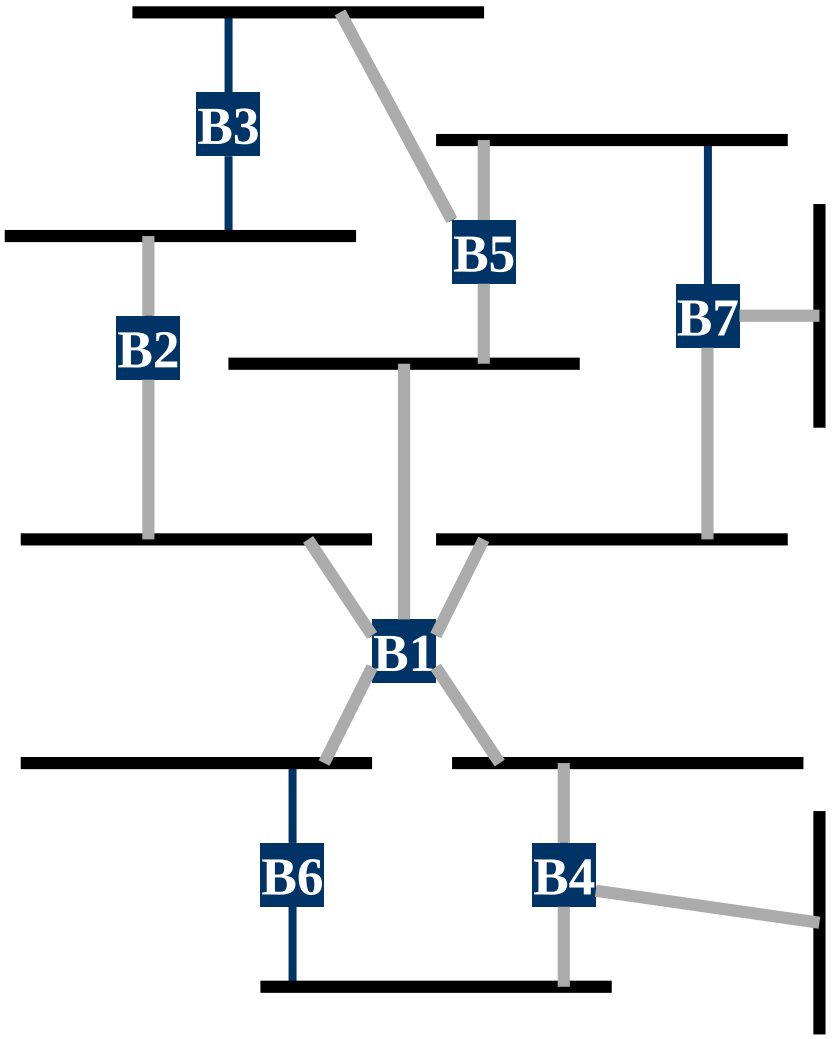
Spanning Tree Algorithm Steps

- Root of the spanning tree is the bridge with the lowest identifier.
  - All ports are part of tree
- Each bridge finds shortest path to the root.
  - Remembers port that is on the shortest path
  - Used to forward packets
- Select for each LAN the designated bridge that has the shortest path to the root.
  - Identifier as tie-breaker
  - Responsible for that LAN



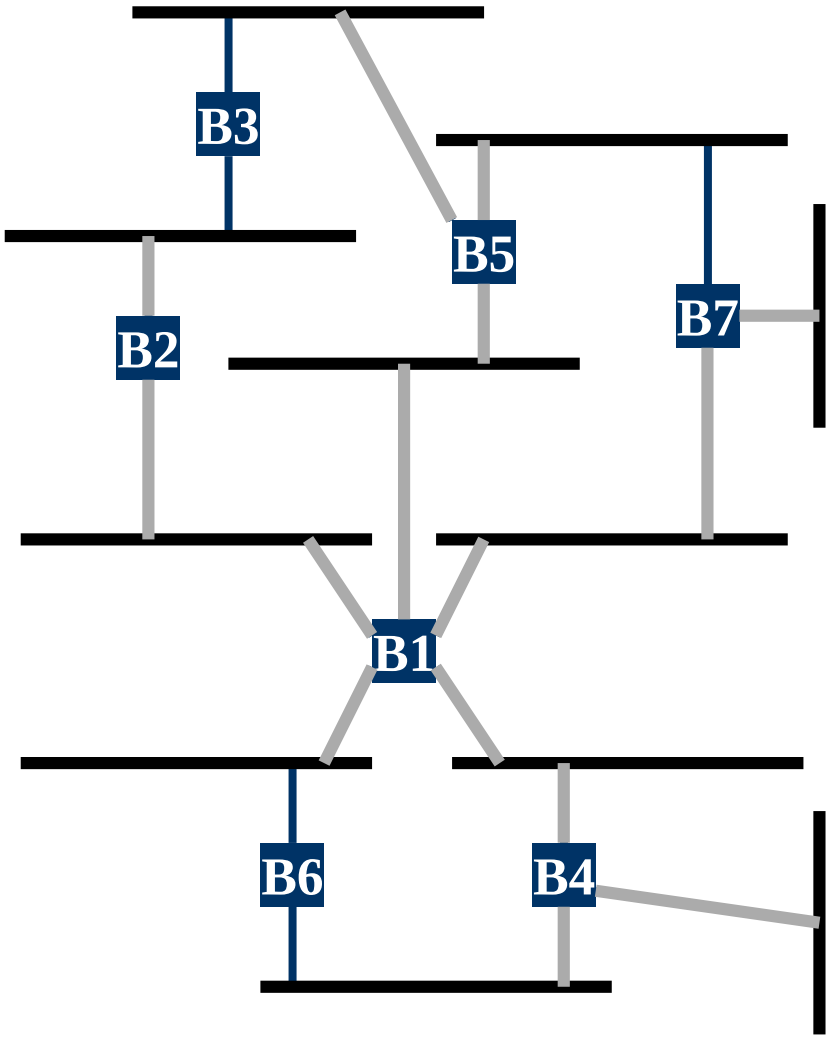
Spanning Tree Algorithm

- Each node sends configuration message to all neighbors.
  - Identifier of the sender
  - Id of the presumed root
  - Distance to the presumed root
  - E.g. B5 sends (B5, B5, 0)
- When B receive a message, it decide whether the solution is better than their local solution.
  - A root with a lower identifier?
  - Same root but lower distance?
  - Same root, distance but sender has lower identifier?
- After convergence, each bridge knows the root, distance to root, root port, and designated bridge for each LAN.



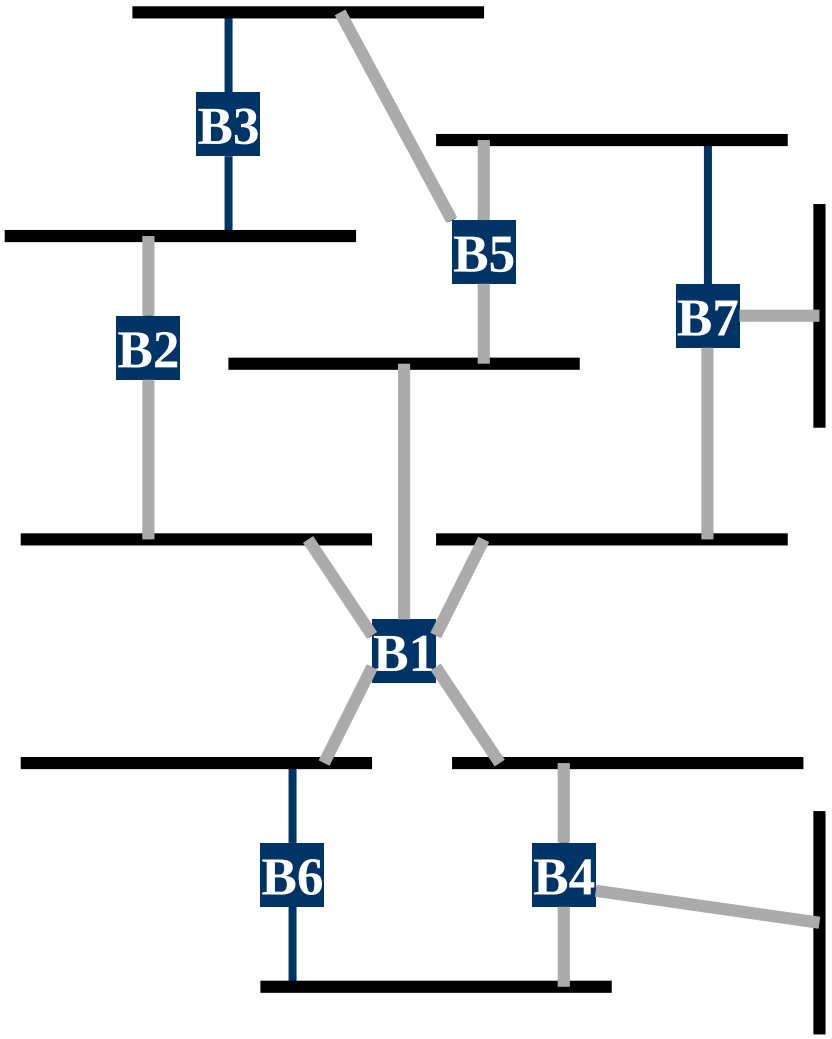
Spanning Tree Algorithm (part 2)

- Each bridge B can now select which of its ports make up the spanning tree:
  - B's root port
  - All ports for which B is the designated bridge on the LAN
- Bridges can not configure their ports.
  - *Forwarding state* or *blocked state*, depending on whether the port is part of the spanning tree
- Root periodically sends configuration messages and bridges forward them over LANs they are responsible for.

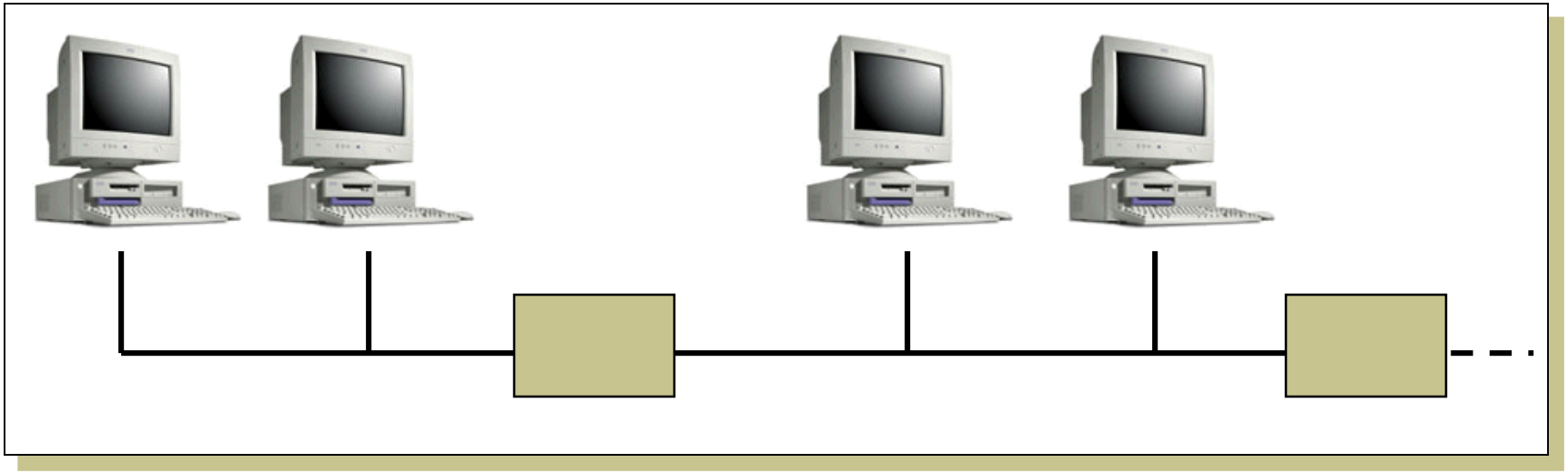


Spanning Tree Algorithm Example

- Node B2:
  - Sends (B2, B2, 0)
  - Receives (B1, B1, 0) from B1
  - Sends (B2, B1, 1) “up”
  - Continues the forwarding forever
- Node B1:
  - Will send notifications forever
- Node B7:
  - Sends (B7, B7, 0)
  - Receives (B1, B1, 0) from B1
  - Sends (B7, B1, 1) “up” and “right”
  - Receives (B5, B5, 0) - ignored
  - Receives (B5, B1, 1) - better
  - Continues forwarding the B1 messages forever to the “right”



Problem 2 – Bridging Weaknesses



- Doesn’t handle incompatible LAN technologies
- How well does it scale?

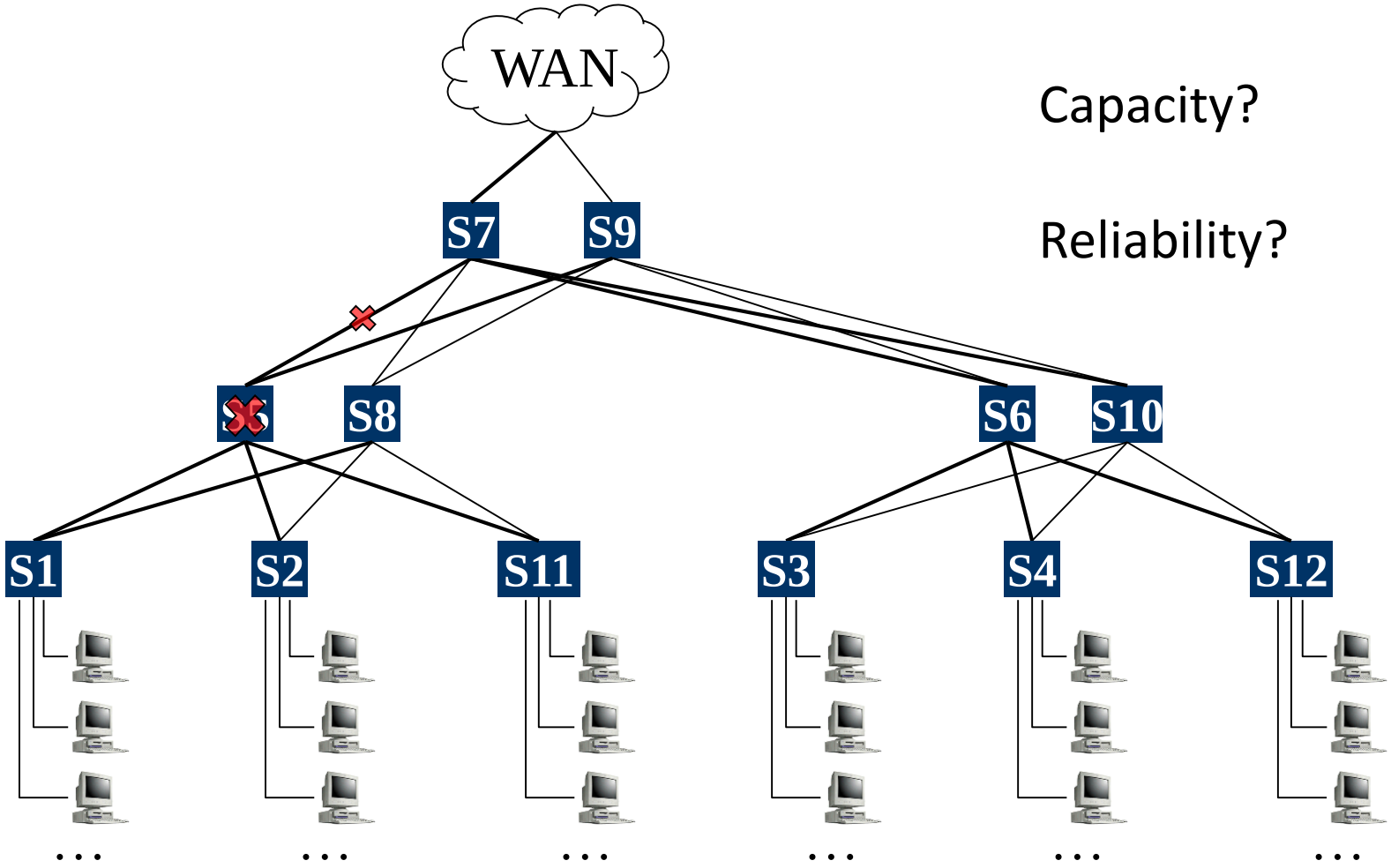
Outline

- Bridging
- Switch Topologies
- Internetworks
  - Methods for packet forwarding
- Traditional IP addressing

“Traditional” Topology

- Hierarchical single tree
- Redundancy for reliability
- Spanning tree (or variant) for loop-free-ness

"Traditional" Topology



Capacity?

Reliability?

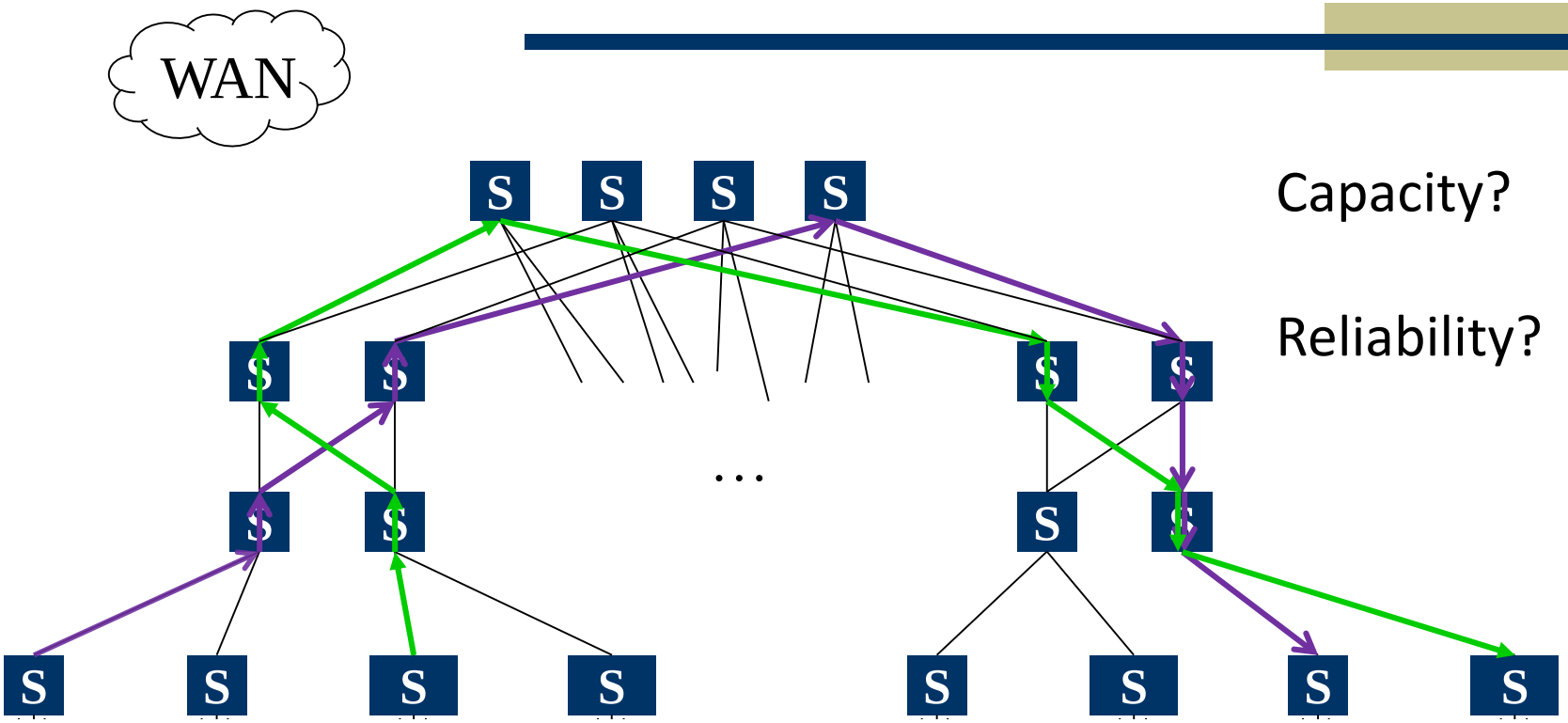
Parallel/HPC “Supercomputer” Topology

- Very high bandwidth & low latency
- Dense connection for performance
- Dedicated resources
- Often local communication
- (e.g. hypercube or torus)

A Data Center Is (Not) A Supercomputer

- Mix of applications sharing resources
- Dynamic provisioning of machines & network
  - Machines that work together may not be next to each other
  - And they might move
- Internal AND external traffic
- Lots of anywhere-to-anywhere bandwidth

“Fat Tree” Topology

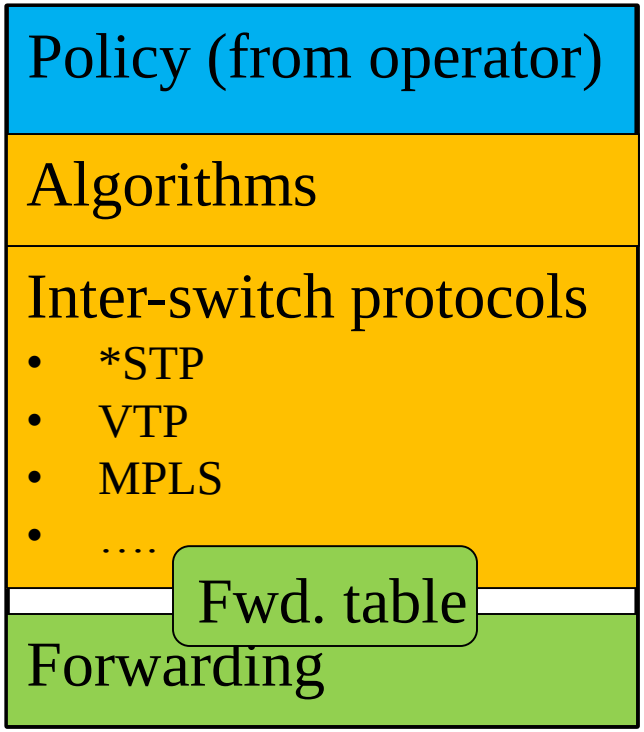


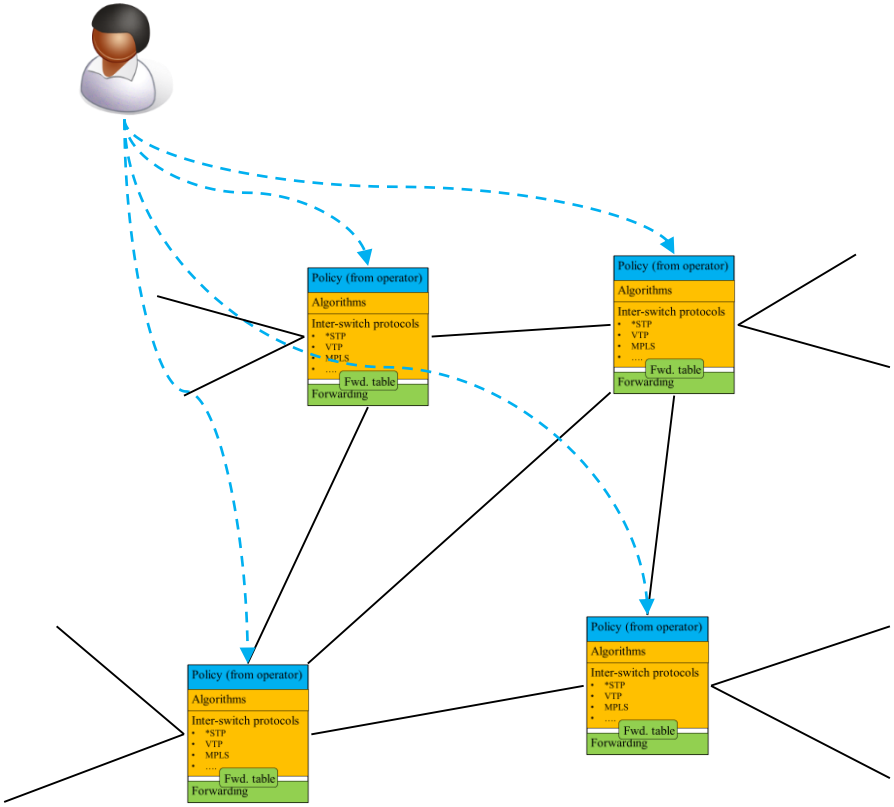
Capacity?

Reliability?

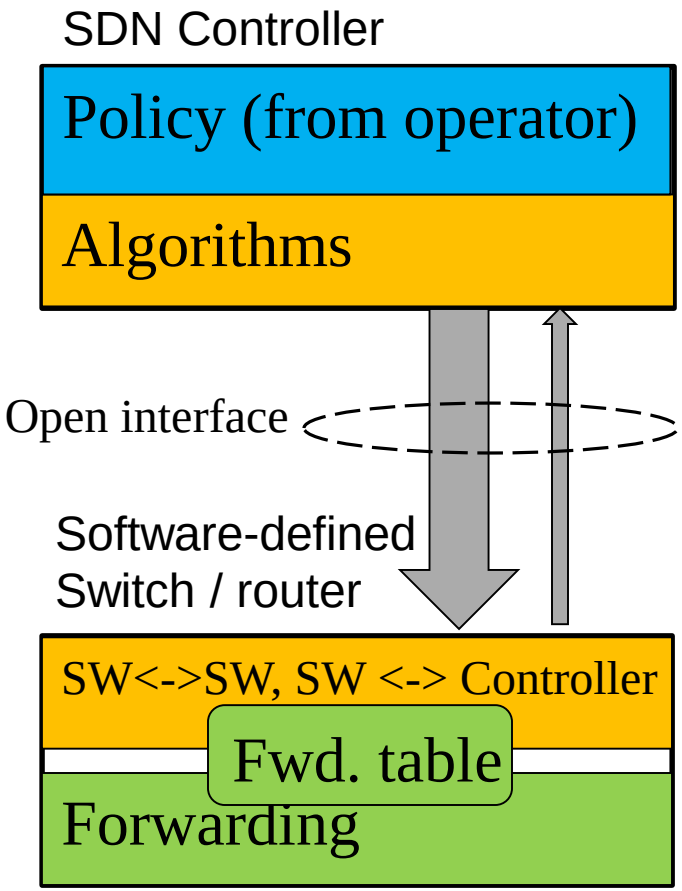
Digression: Software-Defined Networking

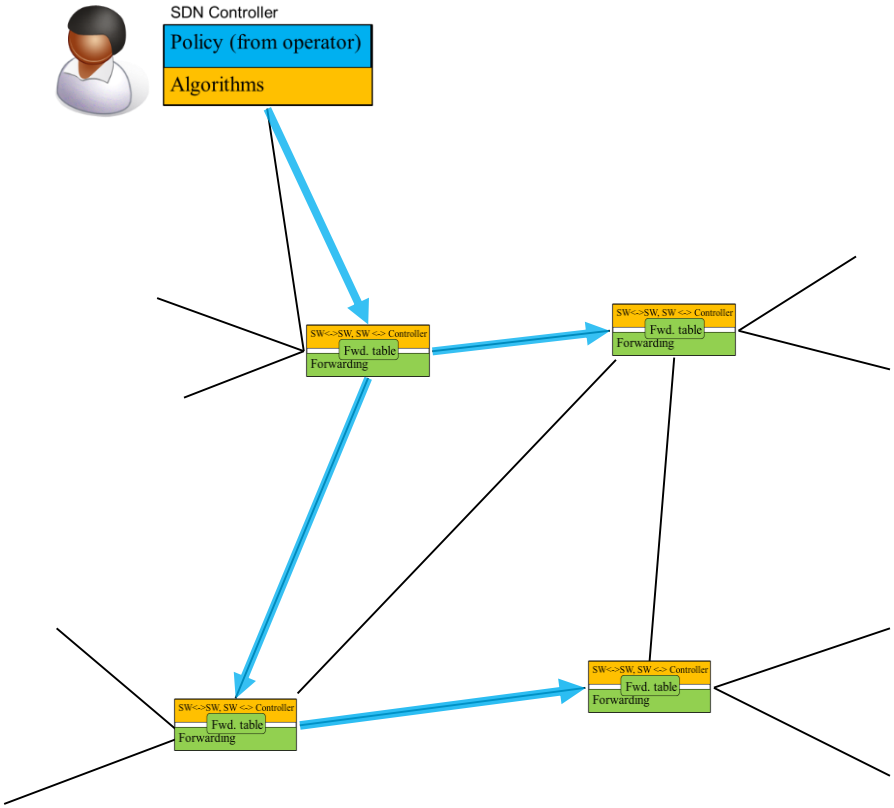
"Vertically-integrated"
Switch / router





Digression: Software-Defined Networking



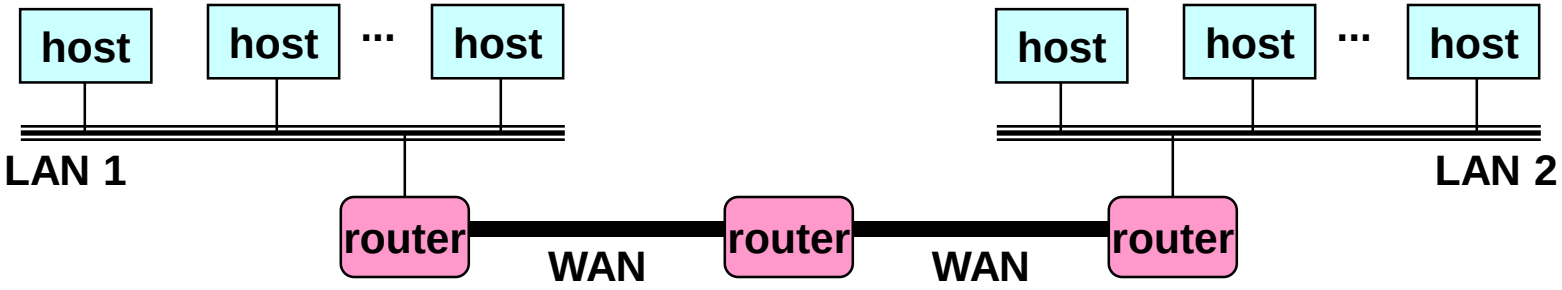


Outline

- Bridging
- Switch Topologies
- Internetworks
  - Methods for packet forwarding
- Traditional IP addressing

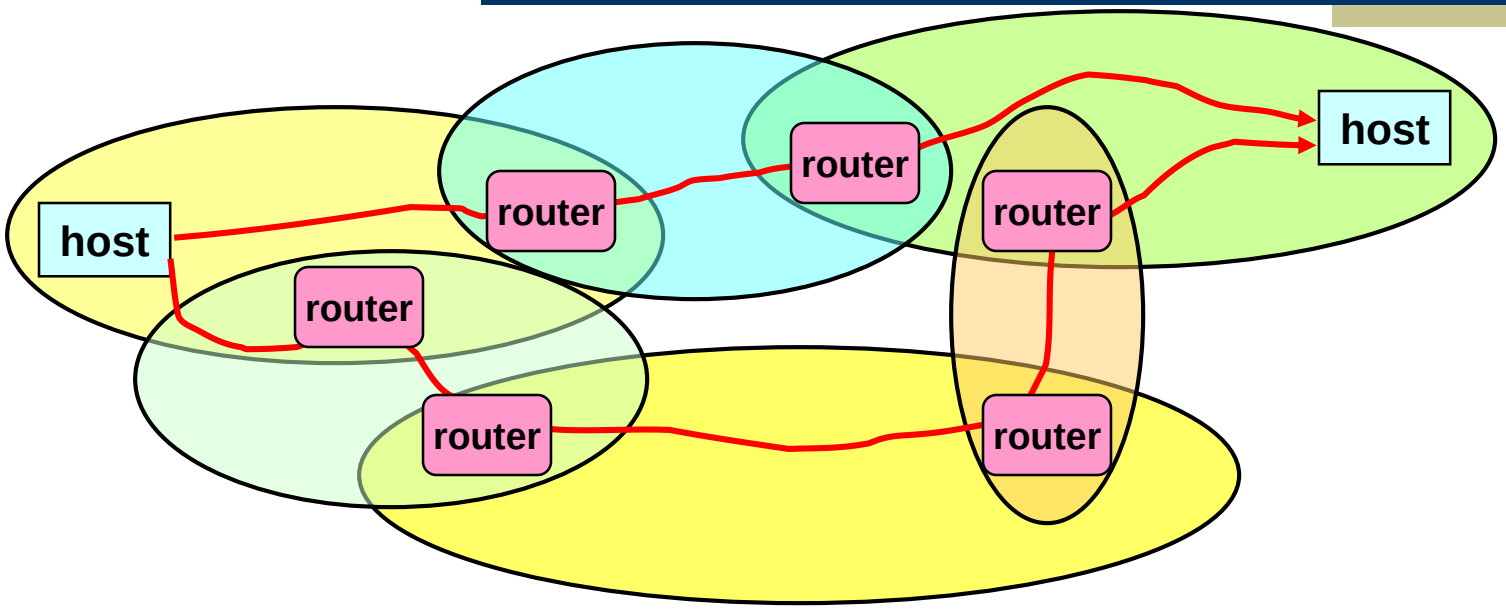
What is an Internetwork?

- Multiple incompatible LANs can be physically connected by specialized computers called *routers*
- The connected networks are called an *internetwork*
  - The "*Internet*" is one (very big & successful) example of an internetwork



LAN 1 and LAN 2 might be completely different, totally incompatible LANs (e.g., Ethernet and ATM)

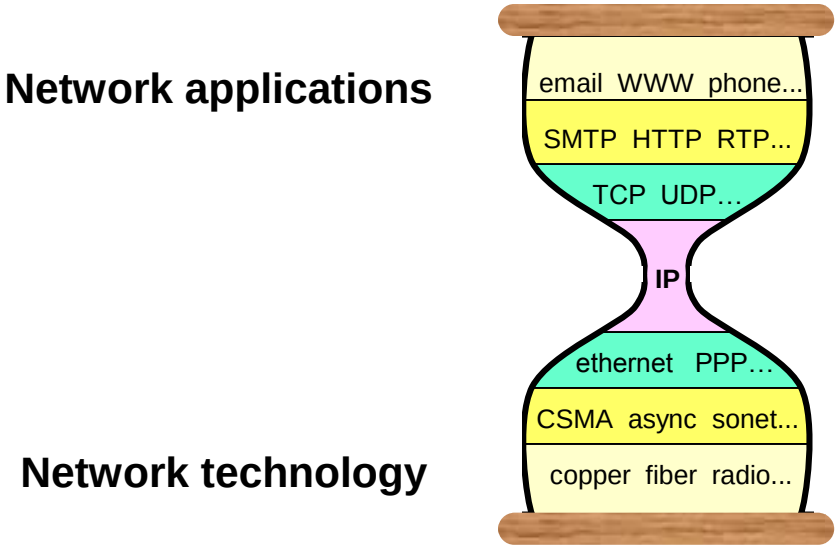
Logical Structure of Internet



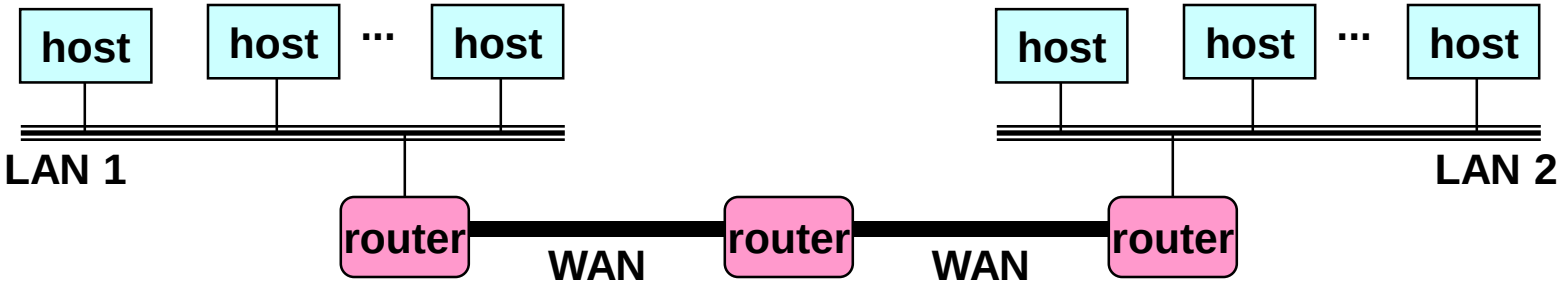
- Ad hoc interconnection of networks
  - No particular topology
  - Vastly different router & link capacities
- Send packets from source to destination by hopping through networks
  - Router connect one network to another
  - Different paths to destination may exist

Internet Protocol (IP)

- Hour Glass Model
  - Create abstraction layer that hides underlying technology from network application software
  - Make as minimal as possible
  - Allows range of current & future technologies
  - Can support many different types of applications



Problem 3: Internetwork Design



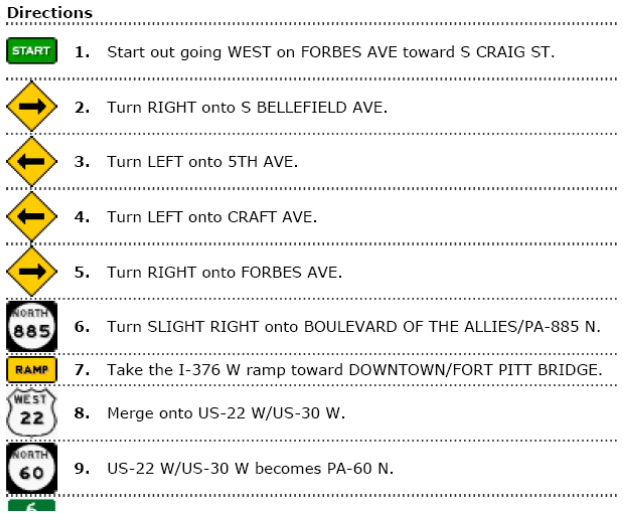
- How do I designate a distant host?
  - Addressing / naming
- How do I send information to a distant host?
  - What gets sent?
  - What route should it take?

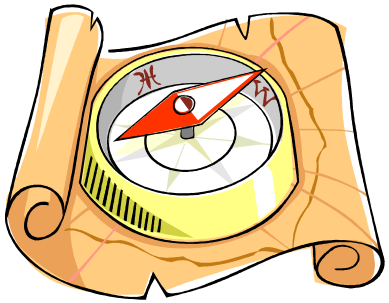
- Must support:
  - Heterogeneity LAN technologies
  - Scalability → ensure ability to grow to worldwide scale

Getting to a Destination

- How do you get driving directions?

- Intersections→routers
- Roads→links/networks
- Roads change slowly





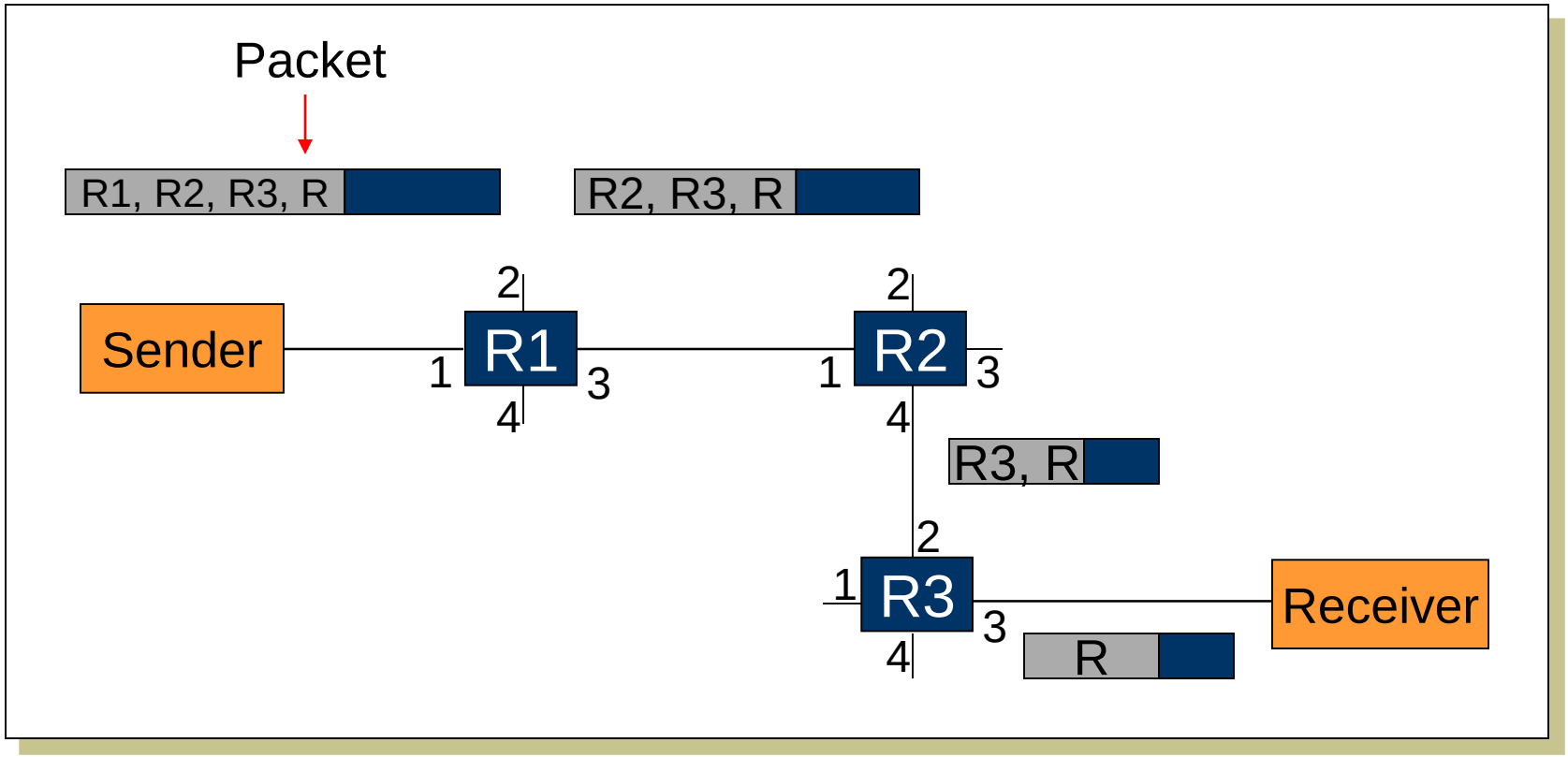
Forwarding Packets

- Table of virtual circuits
  - Connection routed through network to set up state
  - Packets forwarded using connection state
- Source routing
  - Packet carries path
- Table of global addresses (IP)
  - Routers keep next hop for destination
  - Packets carry destination address

Source Routing

- List entire path in packet
  - Driving directions (north 3 hops, east, etc..)
- Router processing
  - Strip first step from packet
  - Examine next step in directions
  - Forward to next step

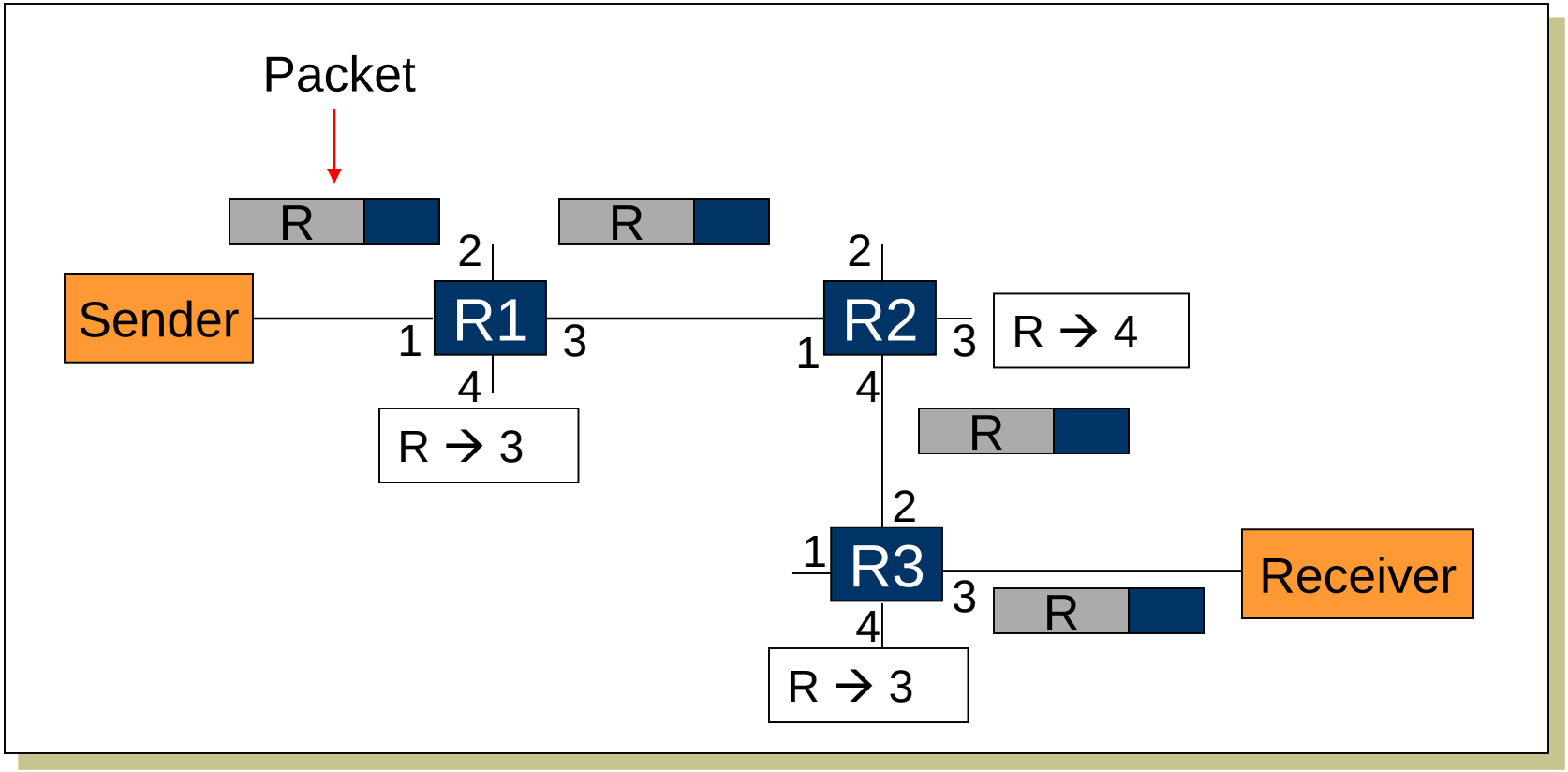
Source Routing Example



Source Routing

- Advantages
  - Switches can be very simple and fast
- Disadvantages
  - Variable (unbounded) header size
  - Sources must know or discover topology (e.g., failures)
- Typical uses
  - Ad-hoc networks (DSR)
  - Machine room networks (Myrinet)

Global Address Example



Global Addresses

- Advantages
  - Stateless – simple error recovery
- Disadvantages
  - Every switch knows about every destination
    - Potentially large tables
  - All packets to destination take same route
  - Need routing protocol to fill table

Comparison

	Source Routing	Global Addresses	Virtual Circuits
Header Size	Worst	OK – Large address	Best
Router Table Size	None	Number of hosts (prefixes)	Number of circuits
Forward Overhead	Best	Prefix matching (Worst)	Pretty Good
Setup Overhead	None	None	Connection Setup
Error Recovery	Tell all hosts	Tell all routers	Tell all routers and Tear down circuit and re-route

Problem 3: Router Table Size

- Global addressing networks (e.g., Internet, Ethernet bridging) require switches/routers to know next hop for all destinations

- How do we avoid large tables?

Outline

- Bridging
- Internetworks
  - Methods for packet forwarding
- Traditional IP addressing

Addressing in IP

- IP addresses are names of interfaces
  - E.g., 128.2.1.1
- Domain Name System (DNS) names are names of hosts
  - E.g., www.cmu.edu
- DNS binds host names to interfaces
- Routing binds interface names to paths

Router Table Size

- One entry for every host on the Internet
  - 630M (1/09) entries,doubling every 2.5 years
- One entry for every LAN
  - Every host on LAN shares prefix
  - Still too many and growing quickly
- One entry for every organization
  - Every host in organization shares prefix
  - Requires careful address allocation

Addressing Considerations

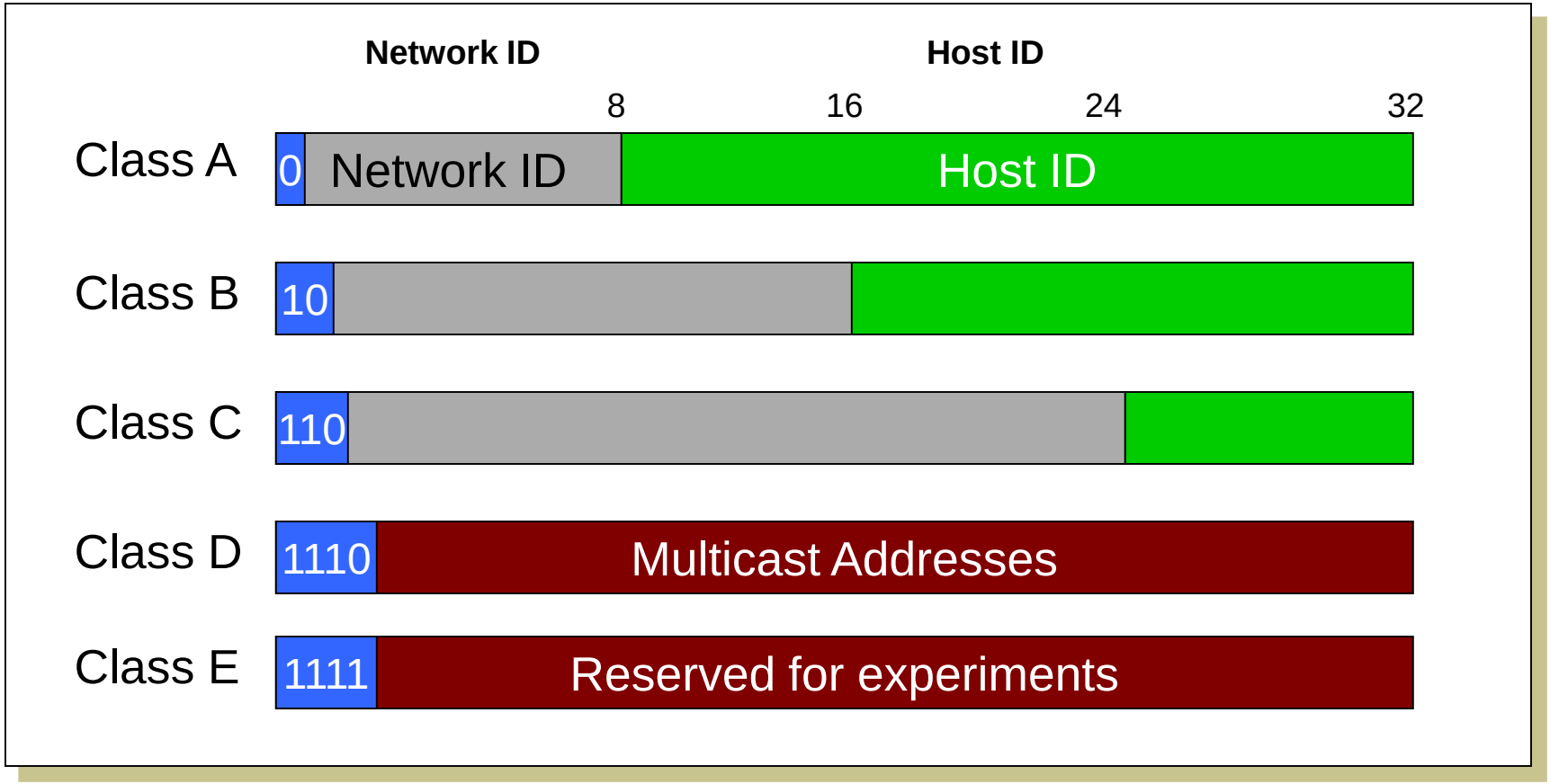
- Hierarchical vs. flat
  - Pennsylvania / Pittsburgh / Oakland / CMU / Steenkiste
    vs.
    Peter Steenkiste: 123-45-6789
    vs.
    Peter Steenkiste: (412)268-0000
- What information would routers need to route to Ethernet addresses?
  - Need hierarchical structure for designing scalable binding from interface name to route!
- What type of Hierarchy?
  - How many levels?
  - Same hierarchy depth for everyone?
  - Same segment size for similar partition?

IP Addresses

- Fixed length: 32 bits
- Initial classful structure (1981) (not relevant now!!!)
- Total IP address size: 4 billion
  - Class A: 128 networks, 16M hosts
  - Class B: 16K networks, 64K hosts
  - Class C: 2M networks, 256 hosts

High Order Bits	Format	Class
0	7 bits of net, 24 bits of host	A
10	14 bits of net, 16 bits of host	B
110	21 bits of net, 8 bits of host	C

IP Address Classes
(Some are Obsolete)



Original IP Route Lookup

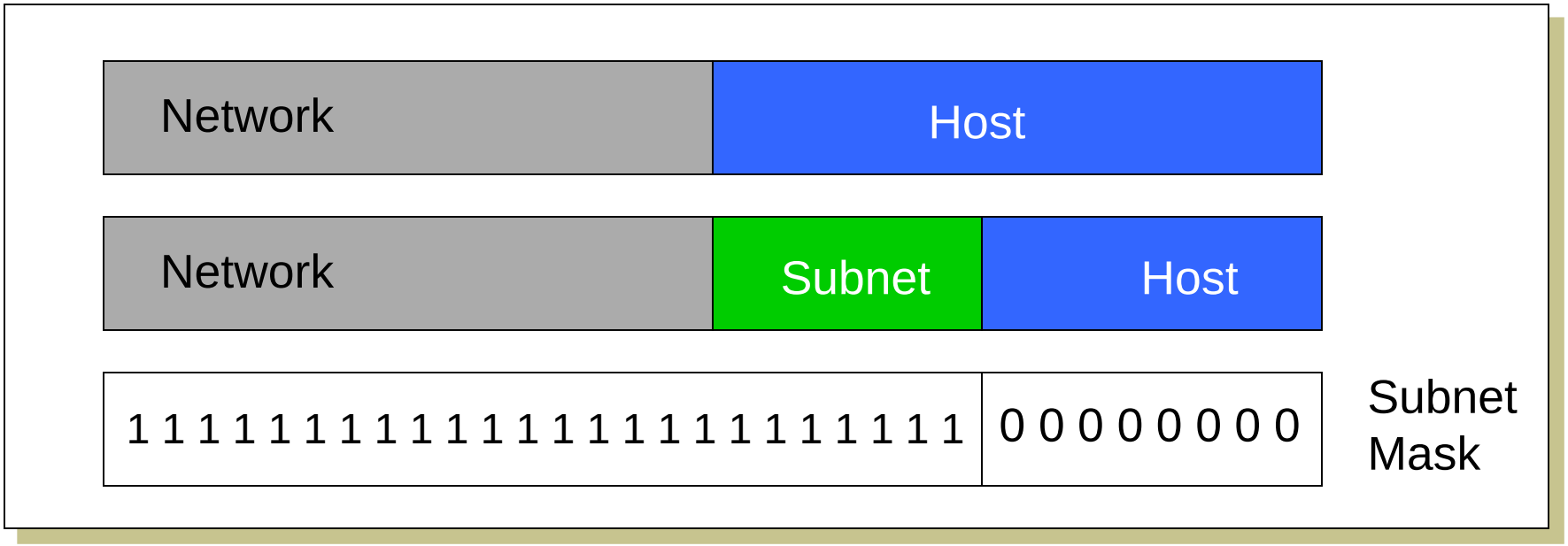
- Address would specify prefix for forwarding table
  - Simple lookup
- www.cmu.edu address 128.2.11.43
  - Class B address – class + network is 128.2
  - Lookup 128.2 in forwarding table
  - Prefix – part of address that really matters for routing
- Forwarding table contains
  - List of class+network entries
  - A few fixed prefix lengths (8/16/24)
- Large tables
  - 2 Million class C networks

Subnet Addressing RFC917 (1984)

- Class A & B networks too big
  - Very few LANs have close to 64K hosts
  - For electrical/LAN limitations, performance or administrative reasons
- Need simple way to get multiple “networks”
  - Use bridging, multiple IP networks or split up single network address ranges (subnet)
- CMU case study in RFC
  - Chose not to adopt – concern that it would not be widely supported ☺

Subnetting

- Add another layer to hierarchy
- Variable length subnet masks
  - Could subnet a class B into several chunks

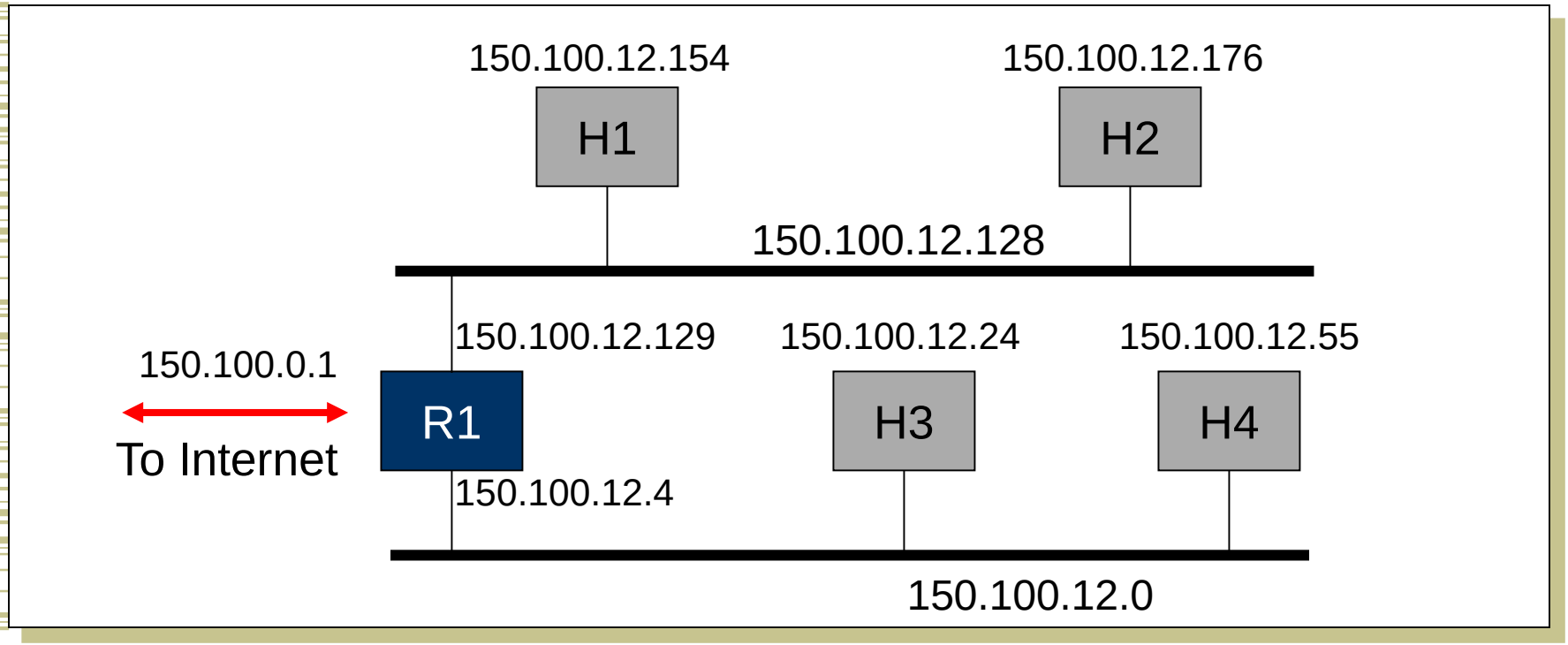


Subnetting Example

- Assume an organization was assigned address 150.100
- Assume < 100 hosts per subnet
- How many host bits do we need?
  - Seven
- What is the network mask?
  - 11111111 11111111 11111111 10000000
  - 255.255.255.128

Forwarding Example

- Assume a packet arrives with address 150.100.12.176
- Step 1: AND address with class + subnet mask



Important Concepts

- Hierarchical addressing critical for scalable system
  - Don’t require everyone to know everyone else
  - Reduces number of updates when something changes